Experiment - 5

Aim

Construct a scheduling program with C that selects the waiting process with the highest priority to execute next.

Algorithm

* Initialize the necessary data structures to store process ID, arrival time, burst time, Priority
* Read no. of processes (N) from the user.
* For each process, read following information.
* Process ID (PID)
* Arrival Time.
* Burst Time (Time required for execution)
* Priority
* Sort the processes based on their arrival time in ascending order.
* Initialize variable current time to 0.
* Initialize a variable completed to 0.
* Create a priority queue or data structures to store processes based on their priority.
* Initialize a variable total_waiting_time to 0.
* While there are still process to execute, repeat the following.
* Calculate the average waiting time as total_waiting_time / N. 15. Print the average waiting time.

Result

The image outlines an algorithm for a Priority scheduling program in C, which selects and executes waiting processes based on their assigned priority level.